

1.4 Basic Vector Calculus

Gradients, Jacobians, and Hessians in the notation this course uses.

1. Gradient

$(J(\boldsymbol{\theta})): \mathbb{R}^I \rightarrow \mathbb{R})$. Vary one coordinate, hold the others. Those partials, stacked, are the gradient. This course writes it as a **row**

$$[\nabla J = \text{Bigl}[\frac{\partial J}{\partial \theta_1} \ ; \ \frac{\partial J}{\partial \theta_2} \ ; \ \cdots \ ; \ \frac{\partial J}{\partial \theta_I} \ \Bigr] \ \text{in} \ \mathbb{R}^{1 \times I}.]$$

Some books write a column. Be consistent inside one derivation. Descent updates look like $(\boldsymbol{\theta}) \leftarrow (\boldsymbol{\theta}) - \alpha (\nabla J)^{\text{top}}$ if (∇J) is a row.

2. Chain rule

If $(J = J(\theta_1(t), \theta_2(t)))$, then

$$[\frac{dJ}{dt} = \frac{\partial J}{\partial \theta_1} \frac{d\theta_1}{dt} + \frac{\partial J}{\partial \theta_2} \frac{d\theta_2}{dt}.]$$

That is a gradient times a Jacobian. Back-propagation (Module 6) is this rule applied to a long composition. You do not need the neural-net story yet. You do need to be willing to write the product.

3. Hessian

If (J) is twice differentiable,

$$[\nabla^2 J = \begin{bmatrix} \partial^2 J / \partial \theta_1^2 & \partial^2 J / \partial \theta_1 \partial \theta_2 \\ \partial^2 J / \partial \theta_2 \partial \theta_1 & \partial^2 J / \partial \theta_2^2 \end{bmatrix}.]$$

The mixed partials match (Schwarz), so the Hessian is **symmetric**. It measures **curvature**.

At a critical point $(\nabla J = \mathbf{0})$:

- $(\nabla^2 J)$ positive definite $(\Rightarrow)$ local **minimum**;
- negative definite $(\Rightarrow)$ local **maximum**;
- indefinite $(\Rightarrow)$ saddle.

Newton's method (week 2) solves a linear system with this matrix every step. Eigenvalues of $(\nabla^2 J)$ are why a long thin valley makes steepest descent zigzag.